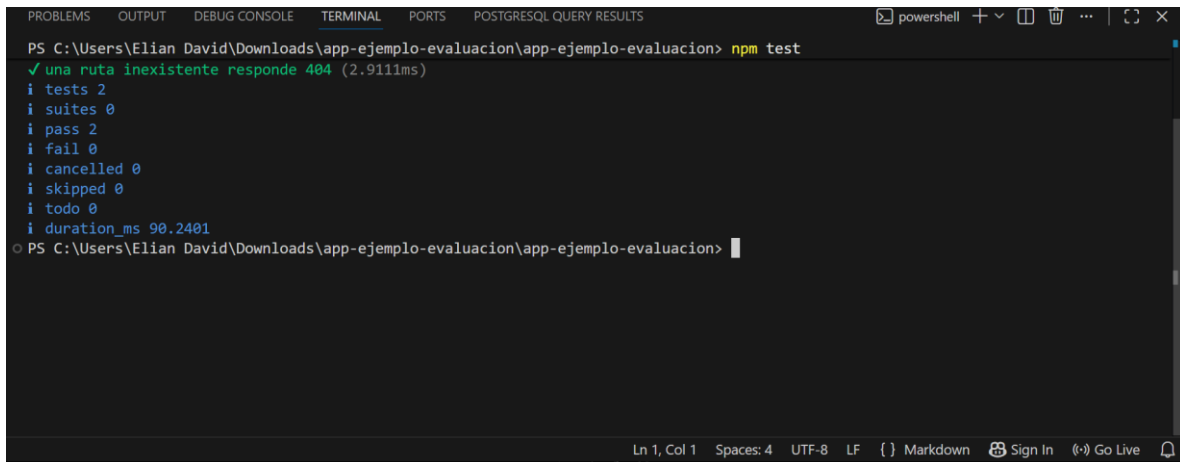


Enlace repositorio

<https://github.com/Davidvillahdz/evaluacion-ejemplo.git>

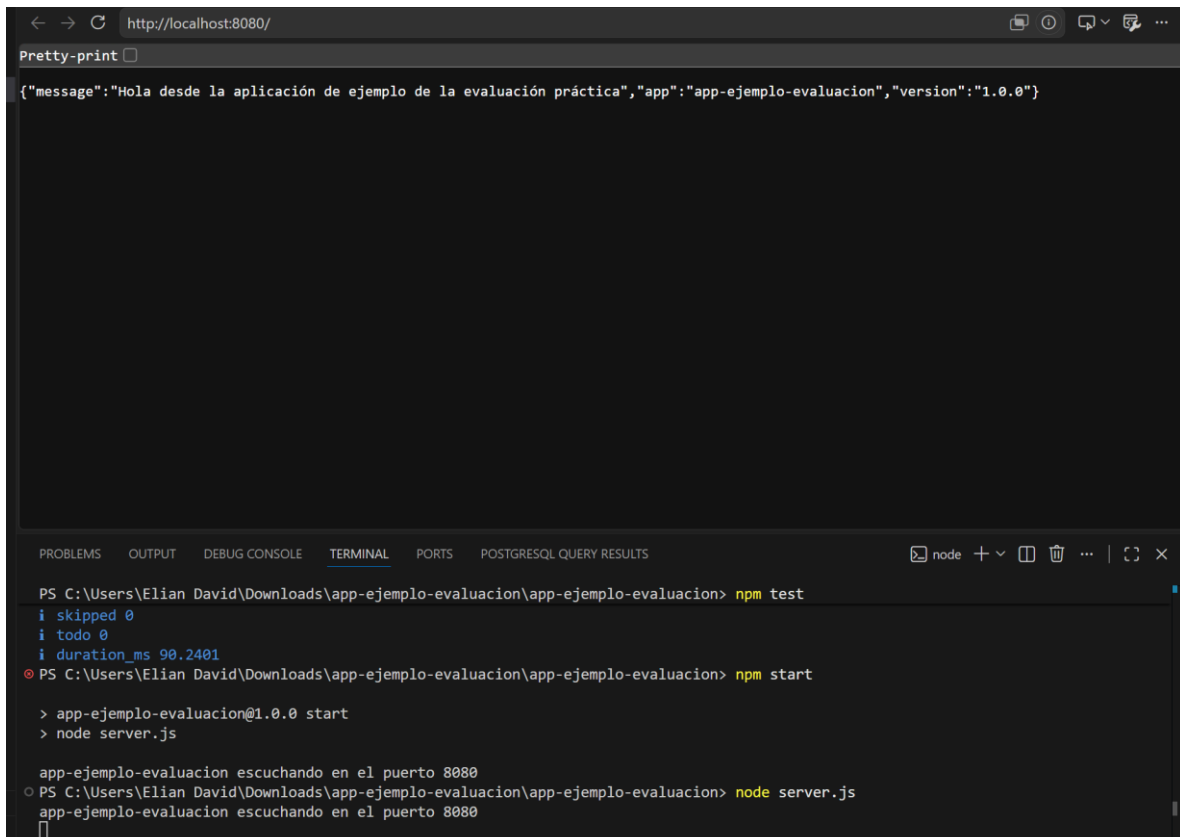
Evidencias capturas

Evidencia del npm test



```
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> npm test
✓ una ruta inexistente responde 404 (2.911ms)
i tests 2
i suites 0
i pass 2
i fail 0
i cancelled 0
i skipped 0
i todo 0
i duration_ms 90.2401
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion>
```

Evidencia de que la aplicación funciona



```
http://localhost:8080/
Pretty-print
{"message":"Hola desde la aplicación de ejemplo de la evaluación práctica","app":"app-ejemplo-evaluacion","version":"1.0.0"}

PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> npm test
i skipped 0
i todo 0
i duration_ms 90.2401
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> npm start
> app-ejemplo-evaluacion@1.0.0 start
> node server.js

app-ejemplo-evaluacion escuchando en el puerto 8080
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> node server.js
app-ejemplo-evaluacion escuchando en el puerto 8080
```

Evidencia de construcción del Dockerfile

```
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> docker build -t app-ejemplo-evaluacion:fallo .
[+] Building 2.3s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 160B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293
=> [internal] load build context
=> => transferring context: 4.30kB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:f9e7bbbd66e2ebaa248e01370132811f9d893b3918793b485344f7684b9da64e
=> => exporting config sha256:77ee3cb81f4cb09a1e46f90d9713c09b4333c0fbf3c62575d4cce87f4b73da9b
=> => exporting attestation manifest sha256:b99d0538a8b41be9f5a3c6823ad1ec2458a0ba723cadbc94191bdd6a8f2af7f5
=> => exporting manifest list sha256:284343260f6688c8654e4597bd26de50fba8744ce26c5f2d99cda4699f715ee4
```

Evidencia de contenedor en ejecución

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion (0.266s)
docker run -d --name app-fallo -p 3001:3000 app-ejemplo-evaluacion:fallo
dabdd3111d1bada2c8e4de17d46d172499284521409a9e0f798d878df51ed281
```

Evidencia de la construcción de la imagen

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
dabdd3111d1b	app-ejemplo-evaluacion:fallo		"docker-entrypoint.s..."	57 seconds ago	Up 56 seconds	0.0.0.0:3001->3000/t
cp, [::]:3001->3000/tcp	app-fallo					
414535a552be	kindest/node:v1.36.1		"/usr/local/bin/entr..."	12 days ago	Up 13 minutes	127.0.0.1:55145->644
3/tcp	reservas-cluster-control-plane					
d81e0d1dad67	kindest/node:v1.36.1		"/usr/local/bin/entr..."	12 days ago	Up 13 minutes	
fc9dfcf325a0	kindest/node:v1.36.1	reservas-cluster-worker				
ee36c338c218	nginx/nginx-prometheus-exporter:1.3	demo-nginx-exporter	"/usr/bin/nginx-prom..."	5 weeks ago	Up 13 minutes	
5c1a6767ad64	grafana/grafana:11.2.0		"/run.sh"	5 weeks ago	Up 13 minutes (healthy)	0.0.0.0:3000->3000/t
cp, [::]:3000->3000/tcp	demo-grafana					
5f1bec8a73d4	mas-alla-del-localhost-api2	demo-api2	"uvicorn app.main:ap..."	5 weeks ago	Up 13 minutes (healthy)	8000/tcp
ca638634e2e5	mas-alla-del-localhost-api1	demo-api1	"uvicorn app.main:ap..."	5 weeks ago	Up 13 minutes (healthy)	8000/tcp
1cb6f4eb2963	oliver006/redis_exporter:v1.63.0	demo-redis-exporter	"/redis_exporter"	5 weeks ago	Up 13 minutes	9121/tcp
d90bc130baa9	redis:7-alpine	demo-redis	"docker-entrypoint.s..."	5 weeks ago	Up 13 minutes (healthy)	6379/tcp

Evidencia de registro de la falla

```
Invoke-RestMethod http://localhost:3001/
Invoke-RestMethod : The underlying connection was closed: The connection was closed unexpectedly.
At line:1 char:1
+ Invoke-RestMethod http://localhost:3001/
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand
```

Evidencia de donde escucha el puerto la falla

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion (0.063s)
docker logs app-fallo

app-ejemplo-evaluacion escuchando en el puerto 8080
```

Evidencia del Dockerfile corregido

```
1  FROM node:20-alpine
2
3  WORKDIR /app
4
5  COPY package*.json ./
6  RUN npm install --omit=dev
7
8  COPY server.js ./
9
10 EXPOSE 8080
11
12 CMD ["node", "server.js"]
```

Evidencia de corrección del error

```
docker build -t app-ejemplo-evaluacion:latest .

[+] Building 1.3s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 192B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  0.4s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293  0.0s
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293  0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 63B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY package*.json ./                             0.0s
=> [4/5] RUN npm install --omit=dev                               0.5s
=> [5/5] COPY server.js ./                                        0.0s
=> exporting to image                                              0.2s
=> => exporting layers                                              0.1s
=> => exporting manifest sha256:b3ab1d8a63cfa7302d823b65caa05f375cbc5b72b93a3313e6d40482db9f7978  0.0s
=> => exporting config sha256:fc6a3bfc156304be3a3a88dd0d6d347c2caa99e52e3e53c6ba22fe2e6be93104  0.0s
=> => exporting attestation manifest sha256:833ff30bff711e79527c239ba64b82b48ac3c1379ea35e8977a50f6177ce2362  0.0s
=> => exporting manifest list sha256:7d105382441a89c36f17eed374e9226ce3aef78fe2cbf302ff315a7380e86536  0.0s
=> => naming to docker.io/library/app-ejemplo-evaluacion:latest  0.0s
=> => unpacking to docker.io/library/app-ejemplo-evaluacion:latest 0.0s
```

Evidencia de que la imagen existe

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion (0.096s)
docker images app-ejemplo-evaluacion
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
app-ejemplo-evaluacion:fallo	284343260f66	193MB	48.4MB	
app-ejemplo-evaluacion:latest	7d105382441a	193MB	48.4MB	

Evidencia de la creaci3n del contenedor

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion (0.291s)
docker run -d --name app-ok -p 8081:8080 app-ejemplo-evaluacion:latest
2e5b7902dfbfbd0ba16941b726c86a74d3410333feac97d71eecd682d16b437b
```

Evidencia de que la aplicaci3n est1 respondiendo correctamente

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion
docker logs app-ok
app-ejemplo-evaluacion escuchando en el puerto 8080

~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion
Invoke-RestMethod http://localhost:8081/
```

message	app	version
Hola desde la aplicaci3n de ejemplo de la evaluaci3n pr1ctica	app-ejemplo-evaluacion	1.0.0

RETO 2.

Evidencia de la creaci3n del manifiesto inicial

```
kubectl apply -f k8s-inicial.yaml
deployment.apps/web-deployment created
service/web-service created
```

Evidencia de los pods corriendo

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion * minikube (0.1s)
kubectl get pods --show-labels
```

NAME	READY	STATUS	RESTARTS	AGE	LABELS
cicd-practica-sd-dddf78bc4-2q8q5	1/1	Running	1 (3m13s ago)	6d22h	app=cicd-practica-sd,pod-template-hash=dddf78bc4
cicd-practica-sd-dddf78bc4-6qd49	1/1	Running	1 (3m13s ago)	6d22h	app=cicd-practica-sd,pod-template-hash=dddf78bc4
cicd-practica-sd-dddf78bc4-wllc7	1/1	Running	1 (3m13s ago)	6d22h	app=cicd-practica-sd,pod-template-hash=dddf78bc4
cicd-practica-sd-dddf78bc4-wwdgv	1/1	Running	1 (3m13s ago)	6d22h	app=cicd-practica-sd,pod-template-hash=dddf78bc4
web-deployment-df5467dfd-4mbp4	1/1	Running	0	73s	app=web,pod-template-hash=df5467dfd
web-deployment-df5467dfd-w5n66	1/1	Running	0	73s	app=web,pod-template-hash=df5467dfd

Evidencia de los dos pods corriendo

```
kubectl get pods -l app=web
```

NAME	READY	STATUS	RESTARTS	AGE
web-deployment-df5467dfd-4mbp4	1/1	Running	0	3m23s
web-deployment-df5467dfd-w5n66	1/1	Running	0	3m23s

Evidencia de archivo sin endopints

```
kubectl get pods -l app=webapp  
No resources found in default namespace.
```

Evidencia de archivo corregido

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: web-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: web  
  template:  
    metadata:  
      labels:  
        app: web  
    spec:  
      containers:  
        - name: web  
          image: app-ejemplo-evaluacion:latest  
          imagePullPolicy: Never  
          ports:  
            - containerPort: 8080  
---  
apiVersion: v1  
kind: Service  
metadata:  
  name: web-service  
spec:  
  selector:  
    app: web  
  ports:  
    - port: 80  
      targetPort: 8080
```

Evidencia del archivo guardado con su configuración

```
kubectl apply -f k8s-inicial.yaml  
deployment.apps/web-deployment unchanged  
service/web-service configured
```

Evidencia de con endpoints

```
kubectl get endpoints web-service  
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice  
NAME           ENDPOINTS                                     AGE  
web-service    10.244.0.172:8080,10.244.0.173:8080         4m26s
```

Evidencia de petición exitosa

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion (0.0.85s)
Invoke-WebRequest http://localhost:8082/

message                                     app                                     version
-----
Hola desde la aplicaci3n de ejemplo de la evaluaci3n pr3ctica app-ejemplo-evaluacion 1.0.0
```

RETO 3.

Evidencia del pipeline inicial

```
github > workflows > ci-cd.yml
1  name: ci-cd
2  on:
3    push:
4      branches: [main]
5
6  jobs:
7    build-test:
8      runs-on: ubuntu-latest
9      steps:
10       - uses: actions/checkout@v4
11       - run: npm ci
12       - run: npm test
13
14    deploy:
15      runs-on: ubuntu-latest
16      steps:
17       - run: docker build -t app:${{ github.sha }} .
18       - run: docker push registry/app:${{ github.sha }}
19       - run: kubectl set image deployment/web-deployment web=registry/app:${{ github.sha }}
```

Evidencia de registro de una prueba que fall3 provocada intencionalmente.

```
app-ejemplo-evaluacion escuchando en el puerto 8080
PS C:\Users\Elian David\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion> npm test

> app-ejemplo-evaluacion@1.0.0 test
> node --test

X GET / responde 200 con un mensaje y una version (21.63ms)
X una ruta inexistente responde 404 (0.7576ms)
```

Evidencia archivo workflow corregido

```

1 name: ci-cd
2
3 on:
4   push:
5     branches: [main]
6
7 jobs:
8   build-test:
9     runs-on: ubuntu-latest
10
11     steps:
12       - uses: actions/checkout@v4
13
14       - run: npm ci
15
16       - run: npm test
17
18   deploy:
19     needs: build-test
20     runs-on: ubuntu-latest
21
22     steps:
23       - uses: actions/checkout@v4
24
25       - run: docker build -t app:${{ github.sha }} .
26
27       - run: echo "Imagen construida correctamente"
28
29       - run: echo "Despliegue ejecutado correctamente"

```

RETO 4

Evidencia de estado inicial del deployment

```

kubectl get deployment web-deployment

```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
web-deployment	2/2	2	2	45m

```

~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion git:(main) 0 • +0 -0
kubectl get pods -l app=web

```

NAME	READY	STATUS	RESTARTS	AGE
web-deployment-df5467dfd-4mbp4	1/1	Running	0	45m
web-deployment-df5467dfd-w5n66	1/1	Running	0	45m

 Check the logs for the web deployment pods. Shift Alt ↵

Evidencia del aumento de las replicas

```
kubectl scale deployment web-deployment --replicas=6
deployment.apps/web-deployment scaled

~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion git:(main) 0 • +0 -0 ✱ minikube (0.082s)
kubectl get deployment web-deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
web-deployment 6/6      6            6           45m

~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion git:(main) 0 • +0 -0
kubectl get pods -l app=web
NAME                                READY   STATUS    RESTARTS   AGE
web-deployment-df5467dfd-4mbp4      1/1     Running   0          45m
web-deployment-df5467dfd-6s86t      1/1     Running   0          8s
web-deployment-df5467dfd-hr5d5      1/1     Running   0          8s
web-deployment-df5467dfd-qd25b      1/1     Running   0          8s
web-deployment-df5467dfd-tqtjw      1/1     Running   0          8s
web-deployment-df5467dfd-w5n66      1/1     Running   0          45m
```

Evidencia del manifiesto y la salida

```
~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion git:(main) 1 • +10 -12 ✱ minikube (0.136s)
kubectl apply -f k8s-inicial.yaml
deployment.apps/web-deployment configured

~\Downloads\app-ejemplo-evaluacion\app-ejemplo-evaluacion git:(main) 1 • +18 -12 ✱ minikube (39.436s)
kubectl rollout status deployment/web-deployment
Waiting for deployment "web-deployment" rollout to finish: 1 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 1 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 1 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 2 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 2 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 2 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 3 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 3 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 3 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 4 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 4 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 4 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 5 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 5 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 5 out of 6 new replicas have been updated...
Waiting for deployment "web-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "web-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "web-deployment" rollout to finish: 1 old replicas are pending termination...
deployment "web-deployment" successfully rolled out

🚧 Check the logs of the newly deployed pods to verify the application is running correctly. Shift Alt ↵
```

```
docker build -t app-ejemplo-evaluacion:v2 .

[+] Building 2.2s (11/11) FINISHED
=> [internal] load build definition from Dockerfile                                docker:desktop-linux 0.0s
=> == transferring dockerfile: 192B                                              0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine                 1.0s
=> [auth] library/node:pull token for registry-1.docker.io                      0.0s
=> [internal] load .dockerignore                                                  0.0s
=> == transferring context: 28                                                    0.0s
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 0.0s
=> == resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6c0fb8363f609372293 0.0s
=> [internal] load build context                                                  0.0s
=> == transferring context: 1.41kB                                                0.0s
=> CACHED [2/5] WORKDIR /app                                                      0.0s
=> [3/5] COPY package*.json ./                                                    0.0s
=> [4/5] RUN npm install --omit=dev                                               0.7s
=> [5/5] COPY server.js ./                                                       0.0s
=> exporting to image                                                            0.3s
=> == exporting layers                                                            0.1s
=> == exporting manifest sha256:d57ec3996e159e0848202cf54578436e9bcdd5f9223f8205af54522e9a55fe2f          0.0s
=> == exporting config sha256:f6124a42b0ecbb2d0c6e4f41bd033e84d7d8a07725b986a32a43682855116bfe          0.0s
=> == exporting attestation manifest sha256:b05790662d2b0dd686756986ea4313e7410e3911b910f90c11f6b197351398764 0.0s
=> == exporting manifest list sha256:85cb1a9f2537dccc999e4b632640b353c010aceb08f0b66a79a48d756ab3e64d3      0.0s
=> == naming to docker.io/library/app-ejemplo-evaluacion:v2                    0.0s
=> == unpacking to docker.io/library/app-ejemplo-evaluacion:v2                 0.0s
```

```
OK 25 - version 1.0.0
OK 26 - version 1.0.0
OK 27 - version 1.0.0
OK 28 - version 1.0.0
OK 29 - version 1.0.0
OK 30 - version 1.0.0
OK 31 - version 1.0.0
OK 32 - version 1.0.0
OK 33 - version 1.0.0
OK 34 - version 1.0.0
OK 35 - version 1.0.0
OK 36 - version 1.0.0
OK 37 - version 1.0.0
OK 38 - version 1.0.0
OK 39 - version 1.0.0
OK 40 - version 1.0.0
OK 41 - version 1.0.0
OK 42 - version 1.0.0
OK 43 - version 1.0.0
OK 44 - version 1.0.0
OK 45 - version 1.0.0
OK 46 - version 1.0.0
OK 47 - version 1.0.0
OK 48 - version 1.0.0
OK 49 - version 1.0.0
OK 50 - version 1.0.0
```